

Ekansh Sahu

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EDUCATION

University of Maryland

College Park, MD

B.S. Major: Computer Science, Minor: Robotics and Automation Systems

Aug. 2023 – May 2027

- **GPA: 3.86** | Dean's List
- **Relevant Coursework:** Algorithms, Robotics Programming, Discrete Structures

Machine Learning Specialization, Coursera (DeepLearning.AI)

- Coursework: Supervised Machine Learning, Advanced Learning Algorithms, Unsupervised Learning & RL

AWS Cloud Practitioner Certified

EXPERIENCE

Research Assistant (Dr. Haizhao Yang)

March 2025 – December 2025

UM Institute of Advanced Computer Studies, UMD

College Park, MD

- Enhanced state-of-the-art LLM fine-tuning and benchmarking (e.g., Qwen2.5, DeepSeek-VL R1) using PEFT methods such as LoRA, leveraging LLaMA-Factory, HuggingFace Transformers, and PyTorch in multi-GPU distributed environments.
- Collaborated with cross-functional research teams to develop custom evaluation pipelines, manage virtualized training workflows on Linux servers, and apply instruction tuning.

Tech Fellow (Mr. Benjamin Wu)

Aug. 2026 – Dec. 2026

Fischell Institute for Biomedical Devices, Nebula Fellowship Program

College Park, MD

- Built a rigorous evaluation pipeline for GPT-5-Nano on scientific plot understanding in dynamic systems, using tick-based spacing normalization.
- Selected and justified set-based similarity metrics to evaluate structural agreement between predicted and ground-truth plots to reveal hidden failure nodes in predictions.

Technology Assistantship Intern (Dr. Raff Viglianti)

Aug. 2025 – Jan. 2026

Maryland Institute of Technology in Humanities

College Park, MD

- Engineered an Astro-based frontend framework with CETEIcean library for efficient client-side parsing and rendering of TEI XML using vanilla JavaScript.
- Implemented TEI Processing Model logic for dynamic, build-time transformation and styling of TEI elements via automated XML processing workflows.

Web Developer and Designer

May 2025 – May 2026

Campus Coders Crew

College Park, MD

- Developed and deployed customer-facing websites from Figma wireframes to production, utilizing HTML, CSS, JavaScript, and React, and integrating API functionalities.
- Collaborated with team members to design, build, and optimize web products ensuring scalability and intuitive interface while supporting backend integrations with MySQL and Postgres.

Academic Peer Mentor (FIRE: Bioinspired Robotics)

Aug. 2024 – Dec. 2024

Office of Undergraduate Research, UMD

College Park, MD

- Mentoring classes of 40 FIRE students by facilitating discussions, aiding team projects, addressing questions, and helping in presenting research and papers.

Resident Assistant

Aug. 2025 – Present

Department of Resident Life, UMD

College Park, MD

- Fostered a safe, inclusive residential community by providing peer support, conflict mediation, and crisis response while upholding university policies.
- Led community-building initiatives and programs for 49 residents to promote engagement and academic success.

Community Assistant

Aug. 2024 – Aug. 2025

Department of Resident Life, UMD

College Park, MD

- Administered key distribution, package processing, and service desk operations for 706 residents.
- First point of contact for visitors; issued core key changes and troubleshooted issues in halls.

Vice President for Diveristy, Equity and Inclusion

Aug. 2023 – May 2024

South Hill Area Council, University of Maryland

College Park, MD

- Managed comprehensive diversity programs and initiatives within the South Hill Council, helping to foster an inclusive campus culture.

PROJECTS

Bioinspired Flight Optimization | *Arduino IDE, CAD, C++* | [Research Poster](#)

- Designed and tested flexible TPU-85A wingtips controlled via Arduino Nano and micro servos to optimize active wingtip actuation, improving flight efficiency by 32%.
- Conducted iterative testing on a foam body, refining performance through servo torque analysis and material selection, achieving flight times of up to 40 seconds.

TrotMate - Smart Roadtrip Planner | *React, Node.js, Google Maps API* | [Github](#)

- Developed a full-stack web app with automated daily stop suggestions by integrating Google Maps, Places, and Directions APIs to segment routes based on custom drive-time limits.
- Implemented an interactive UI for visualizing routes, selecting overnight stays, and exporting trips.

Knowtion - Health Tracker Web App | *Flask, React, Librosa, XGBoost* | [Github](#)

- Real-time mental health app that analyzes audio and physiological signals to detect stress and emotional shifts using datasets like RAVDESS and WESAD.
- Developed a full-stack pipeline integrating ML inference to help individuals monitor mood patterns.

Portfolio Website | *React, GSAP, Framer Motion, Supabase*

- Designed an interactive, retro-themed portfolio featuring draggable dialog boxes, animated transitions, and real-time journal updates.

TECHNICAL SKILLS

Languages: Python, Java, C, JavaScript, SQL (Postgres), HTML/CSS, R

ML & AI: scikit-learn, TensorFlow, Keras, Data Modeling, LoRA, PEFT, HuggingFace

Robotics: Kinematics, Dynamics, Feedback Control, Perception, Sensor Data Processing

Systems & Software: Linux/Unix, Git, Docker, React, Node.js, Flask, FastAPI